Phonologically Conditioned Multiple Feature Mutation in Maskelynes

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1. Main Claim

- Phonologically Conditioned Multiple Feature Mutation (MFM) in Maskelynes ambitranstive verbs involves three different phonological changes.
- I argue for an account based on concatenative affixation of floating [+voice] and [–continuant] features.
- Purely phonological OT constraint *Twin and *G maintain modularity between phonology and morphology.

2. Theoretical Background

- Generalized Nonlinear Affixation (Bermúdez-Otero, 2012): Featural affixes (+v) and (–c) ↔ Ambitr
- Containment Optimality Theory (Prince & Smolensky, 1993): No Deletion
- Colored Contaiment (van Oostendorp, 2007): Morphological affiliation represented as color

(1) RealizeMorpheme (=RM)
Count one violaton for every color that is not affiliated to any phonetically visible material.

3. Maskelynes Phonology

- In Maskelynes (Oceanic, Vanuatu) prenasalization is contrastive in plosives.
- Voicing is still phonologically active, as seen in voicing assimilations and final devoicing.

⇒ Prenasalized stops are [+voice] and [nasal]

(2) Consonant Phoneme inventory of Maskelynes

	Labial	Labio-dorsal	Coronal	Dorsal
Voiceless plosives	**p**	**p^w**	**t**	**k**
Voiced prenasalized plosives	^{m}b	^{m}b^w	**^{n}d**	^{ŋ}g
Nasals	m	m^w	n	ŋ
Fricatives	**β**	**$β^w$**	s	**x**
Liquids		l r		
Semivowels		w	j	

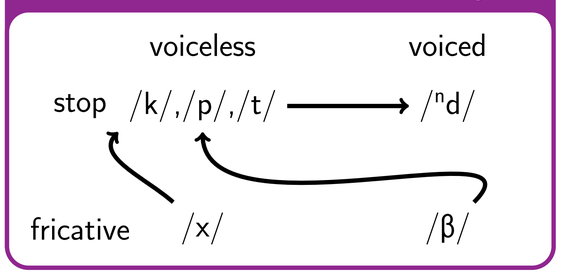
(3) Final devoicing and regressive voicing assimilation

/esi^{m}b/	[esi^{m}p]	'it's finished'
/pu^{n}d/	[py^{n}t]	'hot'
/so^{ŋ}gso^{ŋ}g/	[so^{ŋ}kso^{ŋ}k]	'midden'
/məleβtes/	[məlɛɸtɛs]	'six'

4. Consonant Mutation

- Ambitransitive forms of some verbs are marked by **mutation of the initial consonant** (Healy, 2013).
- Non-coronal fricatives /β/ and /x/ become voiceless plosives /p/ and /k/.
- The voiceless plosive /t/ becomes voiced and prenasalized /^{n}d/.
- No other sounds are changed.

Consonant mutation in Maskelynes



(4) Example of Maskelynes mutation (Healy, 2013, 149-151)

Transitive	Ambitransitive
ti-i	^{n}di
twist-obj	twist\ambitr
xaruβ^w-i	karuβ^w
scratch-obj	scratch\ambitr
βəxas-i	pəxas
annoint-obj	annoint\ambitr

5. Analysis

(5) Evaluation of /(–c)(+v)xaruβ^w/ [karuβ^w]

	*G	RM	Faith
a. (–c)(+v)xaruβ^w		*!	
☞b. (+v)karuβ^w			*
c. (–c)yaruβ^w	*!		*

- A markedness constraint against voiced dorsal sounds (McCarthy & Prince, 1995).

(6) *G: Count one violation for every segmental root node that is phonetically visibly associated to a [DORSAL] feature and a [+voiced] feature.

(7) Evaluation of /(–c)(+v)ti/ [^{n}di]

(–c)(+v) ti	*Twin	RM	Dep(nas)
a. (–c)(+v)ti		*!	
☞b. (–c)^{n}di			*
c. (+v)ti	*!		

- /t/ can only change in voicing, since it is already [–continuant].

(8) *Twin: Count one violation for each segment that is associated to the same feature with the same value twice.

(9) Evaluation of /(–c)(+v)βəxas/ [pəxas]

(–c)(+v)βəxas	*Twin	RM	Dep(nas)	vd→●
a. (–c)(+v)βəxas		*!		*
☞b. (+v)pəxas				**
c. (+v)^{m}bəxas			*!	*
d. (–c)βəxas	*!			*

- /β/ becomes /p/ because inserting a [+nasal] is more costly than a floating [voiced] feature.

(10) Dep(Nas): Count one violation for every colorless (=epenthetic) [nasal] feature

6. Alternative Analyses

- Alternatives often have to make reference to specific morphological categories inside phonology:
  - Indexed constraints (Pater, 2007; Flack, 2007)
  - Cophonology Theory (Orgun, 1996; Inkelas, 1998)
  - Allomorph listing in the lexicon.
- Cophonology Theory and Indexed constraint have to postulate a process of initial voicing and prenasalization.
- An account based on listed stem allomorphs misses the empirical generalizations about the phonological properties of the mutation.
- An account based on phonologically conditioned allomorphy of floating features has to assume more complex representations with up to four different floating features.

7. Discussion

- What about underlying /^{n}g/?

→ Protected by a [nasal]→ • constraint.

- What about mutation of /p/?

✂ No markedness constraint parallel to *G possible.

(11) Hypothetical evaluation of /^{ŋ}g/ [^{ŋ}g]

^{ŋ}g	na→●	*G	...
☞a. ^{ŋ}g		*	
b. k	*!		

8. Conclusion

- Main advantage of the present account: modular and concatenative.
- The seemingly unnatural mutation pattern can be described as the effect of

☞ suffixation of two floating features: [+voice] and [–continuant] and

☞ a markedness constraint against voiced dorsal sounds and

☞ the *Twin constraint.

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